Question ID: 92aa3a94

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

Line k is defined by $y = 7x + \frac{1}{8}$. Line j is perpendicular to line k in the *xy*-plane. What is the slope of line j?

Answer

A. -8

B. $-\frac{1}{7}$

C. $\frac{1}{8}$

D. 7

Correct Answer: B

Rationale

Choice B is correct. It's given that line k is defined by $y = 7x + \frac{1}{8}$. For an equation in slope-intercept form $y = mx + b$, m represents the slope of the line defined by this equation in the *xy*-plane and b represents the *y*-coordinate of the *y*-intercept of this line. Therefore, the slope of line k is 7. It's also given that line j is perpendicular to line k in the *xy*-plane. Therefore, the slope of line j is the opposite reciprocal of the slope of line k. The opposite reciprocal of 7 is $-\frac{1}{7}$. Therefore, the slope of line j is $-\frac{1}{7}$.

Choice A is incorrect. This is the opposite reciprocal of the *y*-coordinate of the *y*-intercept, not the slope, of line k.

Choice C is incorrect. This is the *y*-coordinate of the *y*-intercept of line k, not the slope of line j.

Choice D is incorrect. This is the slope of a line that is parallel, not perpendicular, to line k.